Re-analyzing "I Hope to Hell Nothing Goes Back to The Way It Was Before": COVID-19, Marginalization, and Native Nations

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Abstract

Foxworth et al. (2022) examine the influence of partisanship, along with socio-economic factors, on the spread of COVID-19 among Native American tribes and conclude that tribes in states with Republican governors and where Trump received more votes in the 2016 election fared worse. While their argument and findings sound plausible, the study is flawed methodologically. The most significant issue is that the analysis focuses on the count of cases while not accounting adequately for vast differences in the population of the tribes. Re-analyze the data, I demonstrate that partisanship does appear to be correlated with the spread of Covid-19, but the magnitude of the effect is an order of magnitude smaller than that implied by Foxworth et al.'s (2022) analysis. However, the large proportion of tribes that register no COVID cases cast doubts on those findings as it is unclear whether these represent the actual absence of COVID or issues with the data collection.

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Social Impact & Responsibility

The broader implications of the work presented here is naturally closely related to the that presented in Foxworth et al. (2022). Understanding how the politicization of public health in the context of COVID-19, and how it affects public health outcomes, is quite clearly important. If politicization results in worse public health outcomes, it suggests that it is important to think about how public health policies can be institutionalized in a manner that makes that prioritizes the use of scientific evidence over political incentives. For similar reasons, it is important to understand whether and how minority groups are affect. Foxworth et al.'s (2022) have, thus, identified an important and an interesting question, but their methodology makes their findings easy to dismiss. In short, focusing on the number of cases — rather than the COVID rate — when the sizes of Native American tribes vary enormously, allows the reader to, justifiable, consider their findings meaningless. My reanalysis, demonstrates that their substantive conclusions that politics do matter hold, but that the substantive effects are far smaller than Foxworth et al.'s (2022) findings imply. In addition, their design only considers whether Native Americans in Republican states fare worse than Native Americans in Democratic states, but one might argue that the more relevant question is whether partisan politics affect vulnerable populations more than the general population. That is, Foxworth et al.'s (2022) findings are consistent with an account where Republican states just performed worse in general but there was no difference between the Native and non-Native populations in each state. To address that question, I examine whether the gap in COVID rates between Native and non-Native populations differs depending whether the states is (more or less) Republican or Democratic.

The reanalysis is also important in the current political climate in the United States, where funding for science is being cut and academic freedom is being threatened under the pretext that universities are advancing a 'liberal agenda'. In such circumstances, it is more important than ever for academic research to be scientifically impeachable, especially when it comes to research that has become politically flammable — as research involving partisanship and minority groups can be. Work that is fundamentally flawed and comes to the conclusion that minorities are being advantages, only provides cheap ammunition to those critical of the academia as the work can be cast as prioritizing making a particular point over rigorous analysis. Reanalysis, such as the one presented here, is important, not just to provide scholars with an incentive to think there analysis carefully through, but also for journals to provide for more rigourous peer review.

Foxworth et al. (2022) ask important questions about the role of partisanship in the systematic health inequalities facing Native Americans. They examine whether Covid-19 rates were higher among Native Americans in states that had a Republican Governor and where Trump won more votes in 2016 in addition to other socio-economic factors. The authors conclude that politics matter and note, for example, that case numbers quadruple where there are Republican governors and the effect of high Trump support is similar to the absence of plumbing (p. 10-11).

While interesting, the study has significant limitations that allow the reader a good reason to dismiss the findings. The limitations relate to the operationalization of the dependent variable as the *count of positive Covid cases* through June 11, 2020. This is reasonable if all the Native nations have similar populations. They do not. The smallest Native nation in Foxworth et al.'s (2022) data has five individuals, the largest has 166,395 individuals.[1] In the reanalysis, the rate of positive Covid cases per 100,000 members serves as the dependent variable in an FGLS model. The reanalysis suggests that politics do matter but that the magnitude of the effect is quite different from Foxworth et al.'s (2022).

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[1]Foxworth et al. (2022) propose a solution in an online appendix (https://www.firstnations.org/dataappendix/) but Appendix E shows it is not compelling.

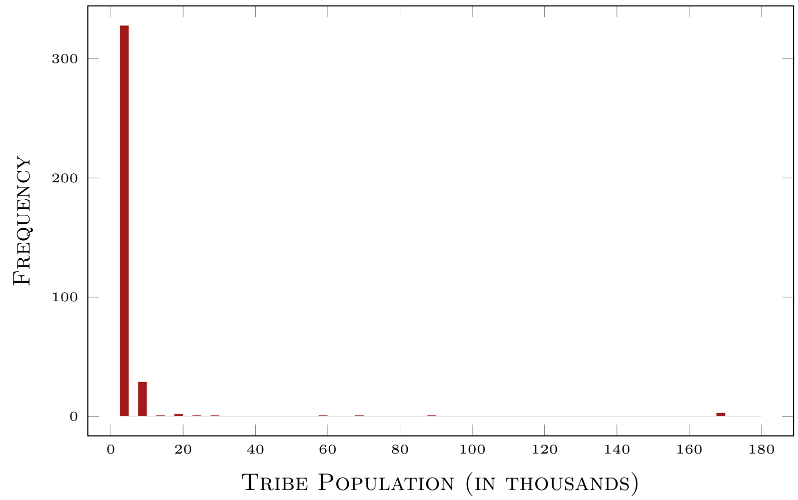
Foxworth et al.'s (2022) design only considers whether Native Americans in Republican states fare worse than Native Americans in Democratic states, but on might argue that the more relevant question is whether partisan politics affect vulnerable populations more than the general population. That is, Foxworth et al.'s (2022) findings are consistent with an account where Republican states just performed worse in general but there was no difference between the Native and non-Native populations in each state. To address that question, I examine whether the gap in COVID rates between Native and non-Native populations differs depending whether the states is (more or less) Republican or Democratic.

I believe the reanalysis is also particularly important in the current political climate in the United States, where funding for science is being cut and academic freedom is being threatened under the pretext that universities are advancing a 'liberal agenda'. In such circumstances, it is more important than ever for academic research to be scientifically impeachable, especially when it comes to research that has become politically flammable — such as research involving partisanship and minority groups. Work that is fundamentally flawed and comes to the conclusion that minorities are being advantages, only provides cheap ammunition to those critical of the academia as the work give the impression making a particular point has precedent over rigorous analysis. Reanalysis, such as the one presented here, is important, not just to provide scholars with an incentive to think there analysis carefully through, but also for journals to provide for more rigourous peer review.

Studying the Spread of Covid-19

Throughout the Covid pandemic there was an active debate about how drastic the efforts to constrain the spread of Covid should be. Many kept a close eye on the spread of Covid across different jurisdictions to gauge the success of different strategies. A basic consideration when comparing jurisdictions is that they vary in size. For example, in early March 2023, the US had recorded over 100 million cases while the Faroe Islands had nearly 35 thousand cases. While the number of cases in the United States is an order of magnitude greater than the number of cases in the Faroe Islands, it is clear that those numbers tell us little about the spread of Covid in the countries because the US population (about 335m) is also an order of magnitude greater than the Faroe Islands (about 50 thousand). In fact, the *Covid-rate* was over twice as high in the Faroe Island.

Figure 1: Distribution of Tribe Population

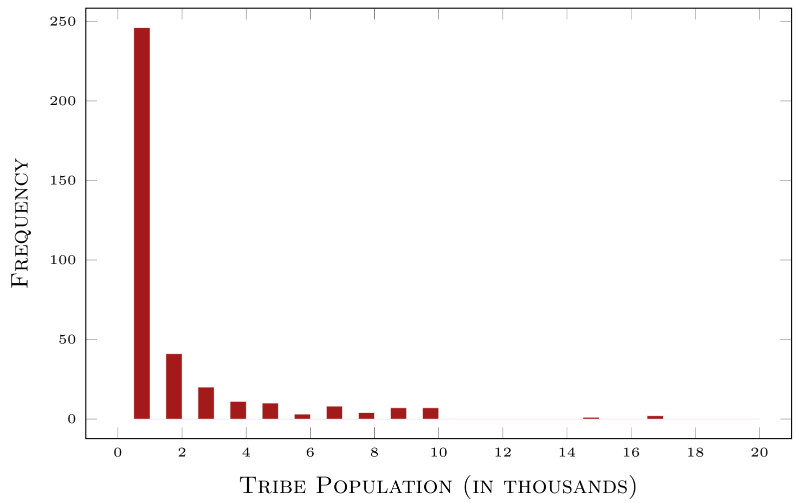


It is difficult to rationalize Foxworth et al.'s (2022) use of the count of Covid cases as being of interest when examining populations that differ in size. While few Covid cases are always better than many, it is hard to imagine circumstances in which success is measured without reference to population size. Simply put, 100 cases in a population of a thousand can hardly be considered an outcome equivalent to 100 cases (and better than, say, 200) in a population of a million. A better way to measure the spread of the disease is to ask what fraction of the population has contracted it. It is an intuitive measure of the authorities' success in containing the disease and one can think of it as the probability that a member of the population has contracted the disease.

The consequences of using the count of Covid-19 cases are minor if the populations are approximately the same. In Foxworth et al.'s (2022) data, that is not the case. Figures 1 and 2 show the population distribution. The first shows all the observations while the second zooms in on those with a population less than 20,000 (about 95% of the tribes). Most of the observations represent small tribes while a handful of tribes are an order of magnitude larger. The population of the four largest tribes accounts for 29% of the native population. Given the variation, the count of cases cannot be considered a reasonable measure of the spread of Covid.

If one accepts that the share of the population that has been infected is a reasonable measure of spread, then analysis considering the number of cases

Figure 2: The Marginal Effect of 2016 Trump Vote
—Tribes with Population Less than 20,000—



as the dependent variable cannot offer useful insights. Suppose that we are interested in assessing the effect of government ideology and that the Faroese government was more left-leaning than the one in the United States. Judging by the count of cases, one would conclude that more right-leaning governments were positively correlated with Covid. However, once population size is taken into account by calculating the Covid-rate, one's conclusions would be reversed. Thus, nothing can be inferred about the effects of covariates (or correlations) when using the count of cases and, thus, the reader is free to reject (or accept) the results. The results provide no information that prompts the reader to update their prior beliefs about whether politics affected the spread of Covid.

It must be noted that Foxworth et al. (2022) do consider the rate of Covid cases in a supplemental appendix. While improving on the results reported in the body of the paper, this analysis also suffers from significant weaknesses related to how the authors deal with heteroscedastic data (see Appendix E). Foxworth et al. (2022) note that the presence of very small tribes means that the Covid rate takes extreme values in some communities and causes heteroscedasticity. That is, the distribution of the residual depends on population size, with the variance being larger when the population size is small. Heteroscedasticity on its own does not result in biased estimates. Thus, regressing the Covid rate on the covariates of interest will provide us with unbiased coefficients. Heteroscedasticity does result in the estimated variances of the coefficients being incorrect, hindering our ability to draw inferences. The OLS estimator is no longer BLUE as it is possible to derive a more efficient estimator.

The textbook approach to address heteroscedasticity involves deriving a more efficient estimator. In practice, each observation is weighted so that the residuals share a common variance using weighted least squares (or feasible generalized least squares when the weights are estimated from the data). Instead of adopting this standard approach, Foxworth et al. (2022) use the log of the Covid rate (cases per 100k inhabitants) as their dependent variable. The problem with this approach is that by taking the log of the dependent variable, the model assumes that the effects of the covariates is proportional to the Covid rate but there is no theoretical reason for thinking that should be the case. Appendix E demonstrates why this is the case and the implications it has in terms of the substantive effects in Foxworth et al.'s (2022) model.

The second limitation of Foxworth et al.'s (2022) analysis concerns the interpretation of the results in terms of substantive effects. It is to Foxworth et al.'s (2022) credit that they do quantify the effects of their covariates as it is important to get a sense of how *much* the political and socio-economic factors matter. However, their modeling choices limit the usefulness of their substantive interpretation. First, the choice to focus on the number of Covid cases makes it difficult to say whether the effects are big or small. For example, their model predicts 2.2 cases for states with a Democratic governor and 8.3 for a Republican governor. In terms of percentages, this is a substantial change. However, whether an increase of 6.1 cases is a substantial change depends on the tribe's population. For a small tribe of 100, this would be quite a large increase. For a large tribe of 100 thousand, it is far less impressive.

Another issue is the use of a non-linear (negative binomial) model as it implies that the effect of a given covariate depends on the values of the other covariates. For example, the effect of having a Republican governor is different for small and large tribes. Foxworth et al. (2022) only report the substantive effect for the median (or mode) of the covariates. The negative binomial model does help a bit as the estimated marginal effects increase as population size increases. However, the model does not capture the relationship that one would expect theoretically, i.e., that the effect doubles as the population doubles. Below I will examine Foxworth et al.'s (2022) estimates of the effects of the political in more detail and give them an interpretation in terms of Covid-rates. I then apply more standard approaches to estimating the effect of the covariates and show that they provide vastly different, but more plausible, estimates of the effects of partisanship.

Reanalysing the Data

Before reanalyzing the data, it is necessary to note a few things about the data. First, out of the 333 observations in Foxworth et al.'s (2022) analysis, 254 have no Covid cases. Although the communities with no Covid cases tend to be smaller, they account for nearly 42% of the total Native population. Having zero Covid cases is, in itself, not bad (in fact, quite the opposite) but the challenge is that we do not know whether these represent actual Covid rates or if these reflect problems with the data collection or lack of reporting. Or, worse, it could be due to a lack of testing, a symptom of the type of systematic inequity in public health provision. If so, the worst outcomes are potentially measured as the best. While the data offers no way of addressing this problem, I estimated models on the subsample with at least one Covid case in addition to models that use the full dataset. Furthermore, in appendix C, I estimate Heckman selection models on the assumption that no Covid cases implies that no testing was available.

Second, some tribes cross state boundaries. Foxworth et al. (2022) address this by conducting their analysis on tribe-state 'dyads'. However, rather than considering the Covid cases within each tribe-state pair (e.g., a separate count of Covid cases for the Navajo nation in Arizona, New Mexico, and Utah), these observations appear to be duplicates — each of the observations for the Navajo nation records 6765 Covid cases and population of 172875. Tribes that cross state boundaries account for 50 of the 333 observations. I estimate models using the full set of observations from Foxworth et al. (2022) and models where I remove the tribes that cross state boundaries.

The appropriate measure of the spread of Covid is the Covid-rate (cases per 100k individuals) and, accordingly, I use this variable as the dependent variable in the reanalysis. To address the heteroscedasticity in the Covid-rate, I estimate feasible generalized least square models. The textbook two-step procedure involves modeling the size of the residuals obtained from OLS and then re-estimating the original model while using the predicted variance to weigh the data.[2] Intuitively, one can think of weighing the data as accounting for the

[2]The code is provided in the appendix. While population size is the main suspect as a source of heteroscedasticity, surprisingly it is not correlated with the size of the residuals. I include the full set of covariates in modeling the variance of the residuals in the original modal and other covariates, such as median age and percentage of English-only households are correlated with the residuals. Foxworth et al.'s (2022) original model includes population on

fact that some of the observations contain less accurate information than other observations. In this case, tribes with small populations might be more likely to have large residuals because a handful of cases can have an outsized effect on the Covid-rate and, if so, smaller tribes have less weight in the estimation.

Foxworth et al. (2022) consider a variety of factors that may influence the Covid-rate, but I will largely restrict my attention to one of their measures of partisanship; Trump's votes share in the state in the 2016 election. Results for the second measure of partisanship — whether the state's governor was a Republican — are presented in appendix A. The other variables are fairly self-explanatory but further details can be found in Foxworth et al. (2022). I consider each of the partisan variables in turn, starting with Trump's vote share in 2016. Table 1, column 1, presents the estimation results of Foxworth et al.'s (2022) negative binomial model where the number of cases is the dependent variable in the first column with the following four columns presenting FGLS models where the dependent variable is the Covid-rate.

The four models (columns 2-5) differ in terms of the observations included. The model in column 2 includes all the observations used by Foxworth et al.'s (2022). The third model excludes tribes that cross state boundaries for the reasons detailed above. The fourth model considers only tribes whose population is less than 20,000 as the distribution of the tribes' population is highly skewed, and the largest tribes may differ in important ways. The cutoff of 20,000 is arbitrary and the goal is simply to explore whether including the largest tribes affects the results. The fifth, and final model, restricts the sample to tribes that reported at least one Covid case.

We can see that Trump's vote share is positively correlated with the spread of Covid much like in Foxworth et al. (2022) with the coefficients being statistically significant at the conventional levels (although only at the 90% level when considering only small tribes). The coefficients in Foxworth et al.'s (2022) model are, of course, not directly comparable with those in models 2-5 because the former is a negative binomial model. It is, however, straightforward to make the results comparable by considering what an increase in Trump's vote share implies in terms of the Covid rate based on the results of Foxworth et al.'s (2022) model. This can be done by dividing the marginal effect of Trump's vote share on the number of Covid cases by the population of the tribal (in 100k).

tribal lands, but not the population of the tribe, as a covariate. As the Covid-rate is based on cases among Native Americans, I also use the size of the tribe, as well as the square of its size, in modeling the variance of the residual from the OLS model.

There are three things to note about this procedure. First, because the negative binomial model produces a marginal effect in terms of the number of cases and the Covid rate is obtained by dividing by population, it implies that the Covid rate depends on the size of the tribe. Second, as Foxworth et al. (2022) uses the total population on tribal lands instead of the tribe's population, an assumption must be made about what share of the total population is a member of the tribe. Different approaches can be applied but here I regress the tribes' populations on the total population and use the predicted tribal population as the denominator in calculating the Covid-rate. The population of the tribe would seem like the more appropriate choice for population size in Foxworth et al.'s (2022) models, which would also avoid the intermediate step of predicting the population of the tribe. Thus, in appendix B, I re-estimate Foxworth et al.'s (2022) negative binomial model where I replace total population with tribal population.[3] Third, the negative binomial model is non-linear, and, thus, the marginal effect of Trump's support depends on population. Thus, to compare the results across models, one first calculates the marginal effect of Trump's vote share conditional on population (with other variables held constant at their mean/mode) and then divides the marginal effect with the (estimated) tribal population.

To compare the two modeling approaches, I plot the marginal effect of Trump's support on the Covid rate at different values of total population. The marginal effect obtained from Foxworth et al.'s (2022) model varies while the marginal effects from the FGLS models do not depend on population size. To avoid cluttering the image, I only plot the smallest and the largest estimates of the marginal effect from the four FGLS models (columns 2-5). I omit the confidence intervals for the same reason as my main concern is to demonstrate that the magnitude of the difference between the two modeling approaches. The figures also show a histogram of population size. Figure 3 displays the marginal effect across the full range of population size. The first thing to note about figure 3 is that the marginal effect implied Foxworth et al.'s (2022) model is always larger than the estimated marginal effect from the FGLS models. The distribution of population size is highly skewed with a handful of very large tribes and, thus, figure 3 is not very informative because there are so few large tribes — the extremely large estimated marginal effects for the largest populations are essentially outliers.

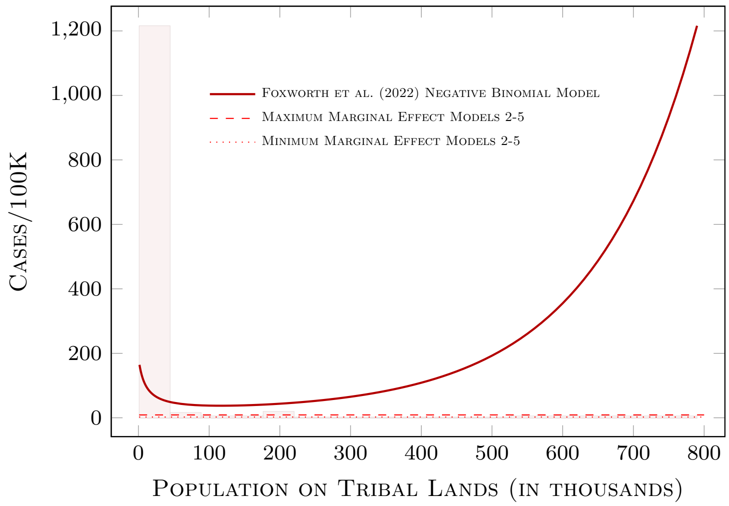
[3]The conclusions regarding the comparison with the FGLS remain unaffected.

Table 1: The Covid Rate & Trump Support in 2016

	(1) FESER†	(2) Orig. Obs.	(3) Unique Obs.	(4) Pop.<20K	(5) >0 cases
% Trump vote in 2016	0.049**	8.890***	7.420***	2.487*	5.343**
	(0.025)	(0.686)	(0.766)	(1.402)	(2.195)
Pop. on Tribal Lands	0.000	0.000	0.000	0.013***	−0.000
	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)
% Native on Tribe's Lands	−3.258**	350.036***	192.107***	190.546***	−185.214
	(1.615)	(36.683)	(58.326)	(54.444)	(338.704)
% English only Households	−5.394***	−930.280***	−409.843***	−304.283***	−441.361
	(0.914)	(50.172)	(120.743)	(111.781)	(402.789)
% Households w/Plumbing	−11.514***	19.800	−535.205**	−766.223***	−375.632
	(4.269)	(290.551)	(264.374)	(263.795)	(1484.365)
Median Age	−0.077*	1.083	−3.747***	−3.193***	−12.428
	(0.043)	(0.944)	(1.172)	(1.046)	(8.603)
Median Household Income	0.000	0.001***	0.001**	0.000	−0.004
	(0.000)	(0.000)	(0.000)	(0.000)	(0.008)
Cases per 100k in State	0.001	−0.003	−0.007	−0.016	−0.067
	(0.001)	(0.007)	(0.008)	(0.015)	(0.096)
Log Casino Sq. Footage	0.275***	13.215***	11.589***	9.777***	7.784
	(0.039)	(2.033)	(2.468)	(2.167)	(17.303)
Rate of Racial Misclassification	−2.387	−237.705***	−92.884	64.642	−232.035
	(1.795)	(48.863)	(63.391)	(51.204)	(278.537)
Health Facility on Tribal Land	3.024***	105.222***	150.145***	65.876**	−137.243
	(0.877)	(21.354)	(29.356)	(33.385)	(94.036)
State Recognized Tribe	0.494	313.335***	236.638***	152.133***	−37.239
	(1.146)	(31.466)	(54.227)	(57.818)	(239.146)
Phoenix Indian Health Board	0.093	−39.747	187.477***	189.709***	254.633
	(0.497)	(27.097)	(55.448)	(50.087)	(266.834)
Constant	14.966**	260.200	461.152*	653.848***	1656.125
	(6.203)	(291.615)	(241.761)	(251.169)	(1203.245)
$\ln(\alpha)$	2.368***				
	(0.133)				
Observations	333	333	283	255	55
Log Likelihood	−454.22	−2448.70	−1994.79	−1794.90	−423.23

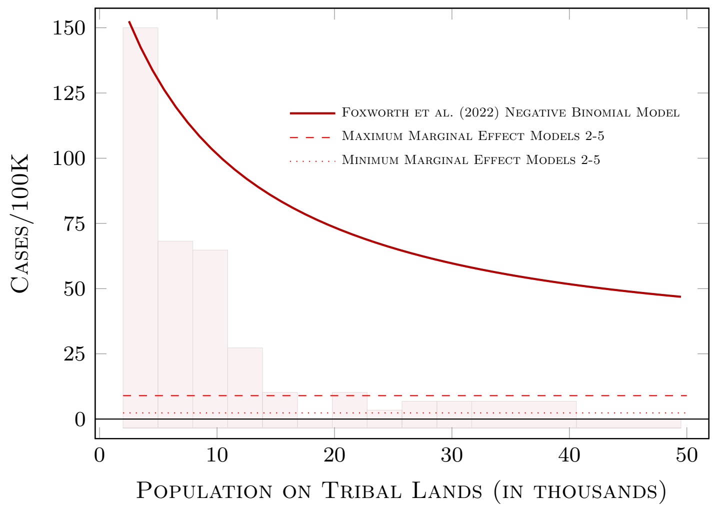
† Foxworth et al. (2022). Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01

Figure 3: The Marginal Effect of 2016 Trump Vote
—Full Range of Population—



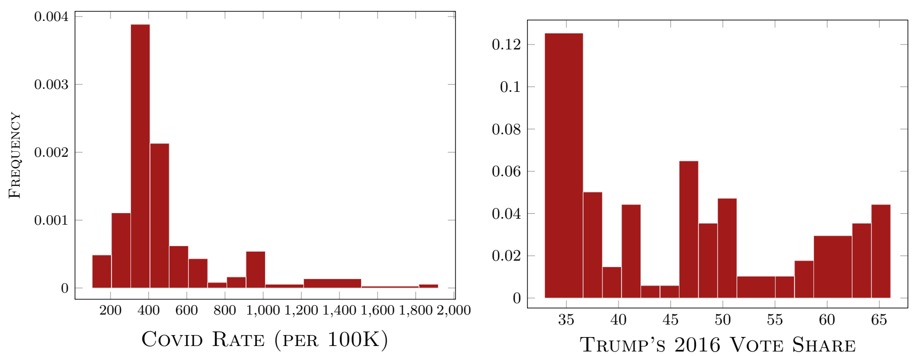
To offer a clearer picture of the estimated marginal effects of Trump's vote share on the Covid rate, figure 4 zooms in on a population range that covers most of the tribes, i.e., tribes with populations less than 50,000. The figure shows that the estimated marginal effect calculated from Foxworth et al.'s (2022) model is an order of magnitude larger than what the FGLS models suggest. The question then is which estimates to believe. There are different ways of approaching that question. First, I have already argued on methodological grounds that using the count of cases as the dependent variable is inappropriate as it equates a single case in a population of 100 with a single case in a population of 100,000. Second, while we know that heteroscedasticity is an issue when using the Covid rate, we also know that OLS in the presence of heteroscedasticity yields unbiased estimates. A third approach to assessing the results is a bit more subjective and simply involves asking whether the estimated effects are plausible.

Figure 4: The Marginal Effect of 2016 Trump Vote
—Total Population Between 2000 & 50000 Only—



To assess whether the effects are plausible, some context is needed. Foxworth et al.'s (2022) data covers the thirty-four states. Figure 5 displays the distribution of the Covid rate and Trump's vote share across the states. The average Covid rate is 430.5 cases per 100k residents and Trump's average vote share is 48.5%. The marginal effects plot in figure 4 corresponds to the effect of a one percentage point change in Trump's vote share — for these smaller tribes, this corresponds to an increase in the Covid rate by at least 50 cases per 100k to as much as 150 per 100k. These are large effects. At the upper end here, this suggests that a one percentage point increase in Trump support corresponds to an increase that is more than one-third of the average Covid rate across the thirty-four states. If we consider a larger but reasonable change in Trump support, for example, by 10% points (the range in the sample is 30% points and the standard deviation is 9.2) then the corresponding change in the Covid rate ranges from 500 to 1500 cases per 100k individuals. Thus, at the low end, it means doubling the average Covid rate while at the high end, the effect is equivalent to quadrupling the average Covid-rate.

Figure 5: Distribution of
Covid Rate (per 100k) & Trump's 2016 Vote Share
—States with Tribal Lands—



Again, whether the magnitude of the effects is considered plausible or not is a subjective judgment. That said, the magnitude of the estimated effect from Foxworth et al. (2022) would seem to stretch the bounds of credulity. In contrast, the estimates from the FGLS models suggest that the marginal effect of Trump support ranges from 2.5 to 8.9 cases per 100k — or, if we consider a 10-point change in Trump support, 25 to 89 cases per 100k. These would appear to be far more plausible estimates although they are still quite large — a ten-point increase in Trump support corresponds to about a 5% to 20% increase in the Covid rate (in a state with the average Covid rate). The upper end of that range may overestimate the effect of Trump support. The smallest estimate is found when only populations less than 20,000 are considered and the magnitude of the effect also drops when we remove observations that appear multiple times because the tribal lands cross state boundaries (compare models 2 and 3).

To summarize, in reanalyzing the models where Trump's support is used as a measure of partisanship, partisanship is found to matter much like Foxworth et al. (2022) conclude. The findings, however, diverge in terms of the substantive effects. Foxworth et al.'s (2022) models imply effects that are an order of magnitude greater than my reanalysis suggests and represent a significant overstatement of the effects of partisanship. The reanalysis, however, does not suggest that the effects of partisanship are small. On the contrary, in substantive terms, the effects remain fairly large although they are nowhere near as large as Foxworth et al.'s (2022) model implies.

In appendix A, I examine the effects of Foxworth et al.'s (2022) second measure of partisanship, Republican governor, which yields the same conclusions.

The implied marginal effects for the Covid rate from Foxworth et al.'s (2022) model are much larger than those obtained from the FGLS models, and the magnitude is difficult to reconcile with the Covid rates observed.

In Appendix B, I report the estimation of models, and accompanying marginal effects plots, identical to table 1 (and table A.1 for governors) except that the models include the population of the tribe among the covariates in place of the total population on the tribal lands. Apart from it being more natural to use the tribal population when modeling the number of cases within the tribe, it has the advantage that no assumption is required about the relationship between the total and tribal population in calculating the effects of partisanship on the Covid rate from Foxworth et al.'s (2022) model.

A minority of the tribes report Covid cases, and after removing tribes whose lands cross state boundaries (the number of cases appear to be reported by tribe and not by tribe-state dyad), only 55 observations remain. It is difficult to determine whether the tribes that had zero Covid cases truly had zero Covid cases or if it is a limitation of the data collection. However, the data availability may be influenced by the same covariates that are used to model the spread of Covid. To explore this possibility, I model both the selection process and the Covid rate using a Heckman selection model (see Appendix C). The result suggests that partisanship is correlated with the reporting of Covid cases — with more Republican states being correlated with cases being reported — while partisanship is not correlated with a higher Covid rate (where at least one case was reported). Ultimately, the Heckman models cannot speak to the question of whether the absence of Covid cases has to do with the actual spread of Covid or with the data collection. However, it does raise important questions about the role of partisanship, i.e., one interpretation is that the spread of Covid is no worse in more Republican areas but that the positive correlation reported in the results above is the result of there being something about Republican states that results in better reporting of Covid cases.

Finally, I consider a different approach to examining public health inequities related to Native Americans. A limitation of Foxworth et al.'s (2022) approach to estimate the effects of partisanship is that there are reasons to think that partisanship does not just affect the spread of Covid among Native Americans but also the general population. If the goal is to estimate whether partisanship affects Native Americans differently, the Covid rate in the state can serve as a benchmark against which to measure how the Native Nations are affected. While Foxworth et al. (2022) do control for the state-level Covid rate, the difference is better captured by using the difference between each tribe's Covid rate and the statewide rate as the dependent variable, i.e., it captures how much worse Native Americans fare compared to the state as a whole.

The results are reported in Appendix D. They suggest Native Americans experience higher Covid rates relative to residents in the state where their tribal lands are. Tables D.6 & D.5 show that the estimated effect is substantial. For example, a ten-point increase in Trump support in the 2016 election corresponds to 62-103 additional Covid cases per 100k inhabitants (depending on which observations are included). Table D.7 presents the results of Heckman selection models that suggest that partisanship is unlikely to affect the Covid rate differential while it matters in the selection part of the model, i.e., whether at least one Covid case was recorded. This is in line with the findings of the selection models discussed above and, thus, raises important questions about how partisanship affects the observed, or measured, Covid rate. That is, is it because Native Americans are more likely to contract Covid in more Republican leaning states? Or is it because, cases are more likely to be detected?

Conclusions

In their article, Foxworth et al. (2022) argue that Native Americans have suffered disproportionality from health epidemics for political and systematic reasons and set out to provide evidence that this was also the case during the Covid-19 pandemic. Their argument is reasonable and there are, for example, good reasons to think that the politicization of pandemic responses would affect the spread of Covid that, in turn, might disproportionally affect Native Americans. Foxworth et al. (2022) present analysis employing novel data to address this question and conclude that partisanship does indeed affect the spread of Covid among the Native American tribes. However, examination of their methodological approach raises significant concerns about their findings. The reason is simple. In their main analysis, Foxworth et al. (2022) use the count of Covid cases as their dependent variable. That might be a reasonable choice if the Native American tribes are similar in size but that is not the case — the smallest tribe consists of five individuals while the largest counts over 160 thousand individuals. When the population of tribes varies so greatly, the number of cases effectively provides no information about the spread of Covid and, thus, drawing inferences about what factors influence the spread of Covid based on the results is simply not possible.

Another related concern is that one is typically not only interested in whether there is a 'statistically significant' relationship between the spread of Covid and the various covariates. Instead, there is substantial interest in learning how much the various factors matter — a concern that is perhaps especially relevant when it comes to devising ways, or policies, to combat the spread of a deadly disease. The focus on the count of Covid cases also gets in the way here for very much the same reason, i.e., it is simply not a reasonable measure of the spread of Covid. For example, if we find that a policy intervention reduces the number of Covid cases by two, it does not provide us with much information about how effective the intervention is. In a population of one hundred, it might well be something worth pursuing while in a population of one hundred thousand, it would seem far less effective.

In this manuscript, I revisit Foxworth et al.'s (2022) analysis with a focus on the effects of partisanship. The same lessons also apply to the other systematic variables considered in their model. My reanalysis changes the focus from the count of Covid cases to the Covid rate (cases per 100k individuals). I also present Foxworth et al.'s (2022) results in terms of the Covid rate and demonstrate that their model implies a much larger effect of partisanship than the reanalysis suggests — and that, in addition, the implied effects are implausible. Finally, I note that Covid rates are only reported for a small share of the tribes, which introduces the possibility of sample selection. To explore that possibility, I estimate Heckman selection models, which suggest that partisanship affects the selection process (i.e., whether at least one Covid case is recorded) but that partisanship does not affect the Covid rate among the tribes that experience at least one case. This introduces some complications when it comes to interpreting the findings as a lack of information about the data-gathering process, i.e., it is not clear whether observations that are recorded as having zero Covid cases actually had no cases or if there actually were cases, but the data were not reported or collected.

References

Foxworth, R., L. E. Evans, G. R. Sanchez, C. Ellenwood, and C. M. Roybal (2022). "I hope to hell nothing goes back to the way it was before": COVID-19, marginalization, and native nations. *Perspectives on Politics 20*(2), 439–456.

Appendix

A The Spread of Covid & Republican Governors

The second proxy for partisanship that Foxworth et al. (2022) consider is whether the governor of the state is Republican or not. The analysis is analogous to when Trump's support was used as a measure of partisanship and is reported in table A.1. The comparison of the effects of partisanship is slightly more straightforward as the partisanship variable is dichotomous. As with Trump's support, I do find that the coefficient for Republican governors is statistically significant much like in Foxworth et al. (2022). However, in addition to our evaluation of the modeling strategy itself, we can again assess Foxworth et al.'s (2022) results by examining what their results imply about the effect on the Covid rate. Figures A.1 and A.2 graph the predicted effect of having a Republican governor, with the second figure zooming in on the tribal lands with smaller populations that account for the vast majority of the observations.

Table A.1: The Covid Rate & Republican Governors

	(1) FESER†	(2) Orig. Obs.	(3) Unique Obs.	(4) Pop.<20K	(5) >0 cases
Republican Governor	1.394***	234.288***	211.358***	105.943***	179.584***
	(0.481)	(26.131)	(29.321)	(34.499)	(51.081)
Pop. on Tribal Lands	0.000	0.000	0.000	0.007*	−0.001
	(0.000)	(0.000)	(0.000)	(0.004)	(0.001)
% Native on Tribe's Lands	−2.181	361.674***	201.815***	176.542***	−619.972
	(1.669)	(40.136)	(47.009)	(34.452)	(886.533)
% English only Households	−5.058***	−975.421***	−392.599***	−261.990***	−654.772
	(0.881)	(51.249)	(111.068)	(94.874)	(612.196)
% Households w/Plumbing	−11.566**	−166.003	−379.434	−525.101**	−647.806
	(5.711)	(341.487)	(269.800)	(245.758)	(1310.965)
Median Age	−0.043	1.833*	−0.036	−0.424	−5.715
	(0.042)	(0.989)	(0.881)	(0.814)	(14.013)
Median Household Income	0.000	0.001***	0.001***	0.001***	−0.010
	(0.000)	(0.000)	(0.000)	(0.000)	(0.015)
Cases per 100k in State	0.001	−0.007	−0.032	0.008	−0.014
	(0.001)	(0.015)	(0.021)	(0.012)	(0.096)
Log Casino Sq. Footage	0.270***	14.140***	11.748***	7.727***	7.671
	(0.042)	(2.503)	(2.513)	(2.068)	(16.889)
Rate of Racial Misclassification	−2.423	−189.616***	−59.875*	−34.038	−221.311
	(1.728)	(41.406)	(34.557)	(23.575)	(387.973)
Health Facility on Tribal Land	3.009***	114.746***	108.170***	42.371	−26.893
	(0.826)	(27.836)	(32.204)	(28.726)	(114.287)
State Recognized Tribe	1.333	398.936***	290.867***	138.790***	89.357
	(1.136)	(55.010)	(47.921)	(45.843)	(335.372)
Phoenix Indian Health Board	−0.535	−207.243***	144.929**	246.546***	128.253
	(0.514)	(31.908)	(72.274)	(72.990)	(349.906)
Constant	15.201**	711.684**	423.461*	523.851**	2413.724
	(6.412)	(319.216)	(228.627)	(227.522)	(2309.033)
$\ln(\alpha)$	2.341***				
	(0.137)				
Observations	333	333	283	255	55
Log Likelihood	−452.93	−2488.90	−2013.90	−1807.46	−436.51

† Foxworth et al. (2022). Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure A.1: The Effect of Having a Republican Governor
—Full Range of Population—

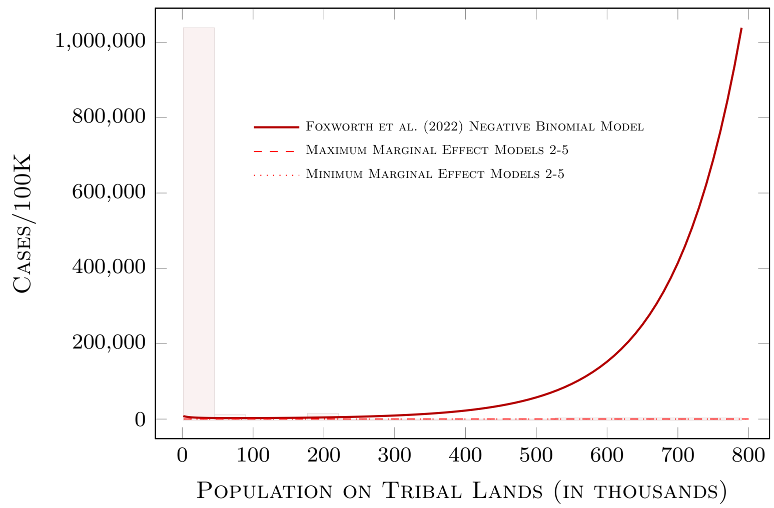
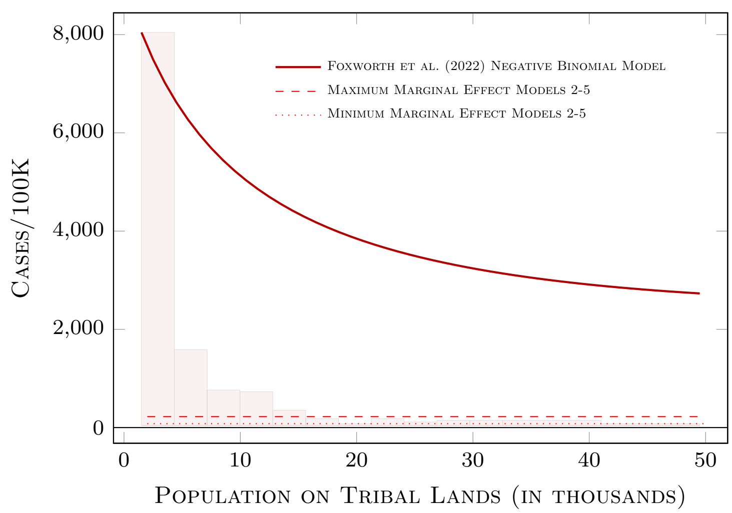


Figure A.2: The Effect of Having a Republican Governor
—Total Population Between 2000 & 50000 Only—



The figures show a pattern that is very similar to the one we saw when examining the effect of Trump's support. That is, the estimated effect is an order of magnitude larger than the effect obtained in the reanalysis using FGLS models. The difference is, however, even more extreme than before and, in fact, leads to nonsensical estimates. For example, for tribal lands with a population of a little over half a million, the estimated effect of having a Republican governor exceeds 100,000 cases (per 100K). Granted, there are only three tribal lands whose population exceeds half a million, i.e., these are outliers in the data, but it is nevertheless an indication that the modeling strategy is suspect.

Focusing on the tribal lands whose population is more typical in figure A.2, the estimated effect of Republican governors is nowhere near as large. The effect on the Covid rate implied by Foxworth et al.'s (2022) is, nevertheless, much larger than the corresponding effect obtained in the FGLS models — in fact, between 11 and 34 times across the range of population size shown in the figure. Foxworth et al.'s (2022) model implies that the Covid rate increases by between 2700 cases per 100K to 8000 cases per 100K. These are very large when viewed in the context of the actual case rates shown in figure 5 — the effect of Republican governors on the case rate is estimated to be larger than the Covid rate in the state with the highest Covid rate in the sample.

The FGLS models shown in columns 2-5 in table A.1 estimate the effect of a Republican governor to range from 106 to 234 cases per 100k individuals. Again, this is a rather large effect — the average Covid rate in the 34 states was 430 cases per 100k individuals. For a state with the average Covid rate, the effect corresponds to about a 25% to 54% increase. A comparison of the four models suggests that the larger estimates may be driven by outliers in the data but even if we discount those estimates, the smallest estimated effect is, in fact, quite large.

On the whole, the reanalysis tells the same story regardless of whether partisanship is measured by Trump's vote share or the governor's party; partisanship is found to be correlated with higher Covid rates. The effect is, however, far smaller than the effect implied by Foxworth et al.'s (2022) models. That does not imply that the magnitude is insignificant. On the contrary, the magnitude of the effect is substantial, suggesting that partisanship has both a statistically and substantively significant effect on the spread of Covid.

B Tribal Population instead of Total Population

Foxworth et al.'s (2022) models of the count of Covid cases include the total population on tribal lands among the covariates. The inclusion of the variable is presumably to account for larger populations being likely to have more Covid cases. However, as the count is a count of Covid cases among members of the tribe, controlling for the total population on tribal lands seems like an odd choice as the number of cases among members of the tribe ought to increase with the size of the tribe but not the size of the population on tribal lands.[4] Thus, in this section I re-estimate the models, replacing total population on tribal lands with the tribal population. Tables B.2-B.3 and figures B.3-B.6 demonstrate that it makes little difference which population variable is used.

[4]It should be noted that the description of the data does not make it clear that the count is a count of cases among Native Americans although the description of the data collection procedures can be taken to suggest that is the case. Given that possibility, it is reasonable to examine the same models where the population of the tribes is included among covariates.

Table B.2: The Covid Rate & Trump Support in 2016
—Controlling for Tribal Population—

	(1) Neg. Binom.	(2) Orig. Obs.	(3) Unique Obs.	(4) Pop.<20K	(5) >0 cases
% Trump vote in 2016	0.056^{**}	4.025^{***}	3.555^{***}	1.186	5.133^{**}
	(0.026)	(0.754)	(1.320)	(0.934)	(2.250)
Tribal Population	0.000	0.020^{***}	0.006	0.059^{***}	−0.000
	(0.000)	(0.000)	(0.004)	(0.007)	(0.002)
% Native on Tribe's Lands	-3.551^{**}	265.403^{***}	180.162^{***}	123.805^{***}	−134.113
	(1.529)	(39.082)	(46.487)	(39.097)	(259.976)
% English only Households	-5.590^{***}	-725.661^{***}	-378.279^{***}	-218.383^{**}	−370.349
	(0.914)	(62.653)	(120.850)	(106.382)	(390.132)
% Households w/Plumbing	-8.999^{*}	−15.238	-564.593^{*}	51.344	197.344
	(5.227)	(302.004)	(334.116)	(228.585)	(1699.253)
Median Age	-0.073^{*}	1.412^{**}	-2.595^{***}	−0.030	−10.364
	(0.043)	(0.606)	(0.942)	(0.622)	(10.024)
Median Household Income	0.000	0.001^{***}	0.002^{***}	0.001^{***}	−0.005
	(0.000)	(0.000)	(0.000)	(0.000)	(0.008)
Cases per 100k in State	0.001	-0.019^{**}	−0.003	-0.055^{***}	−0.091
	(0.001)	(0.008)	(0.007)	(0.010)	(0.096)
Log Casino Sq. Footage	0.267^{***}	11.206^{***}	13.335^{***}	5.787^{***}	1.449
	(0.042)	(2.048)	(1.544)	(1.196)	(19.091)
Rate of Racial Misclassification	−2.140	-154.827^{***}	-120.364^{**}	58.814	−180.850
	(1.623)	(43.339)	(48.866)	(40.360)	(258.662)
Health Facility on Tribal Land	3.088^{***}	117.505^{***}	159.770^{***}	48.512	−113.736
	(0.889)	(31.630)	(44.912)	(30.240)	(105.619)
State Recognized Tribe	1.234	296.353^{***}	321.934^{***}	139.351^{***}	−127.439
	(1.205)	(43.171)	(49.258)	(47.531)	(284.572)
Phoenix Indian Health Board	0.086	55.790	244.754^{***}	114.429^{***}	357.271
	(0.492)	(35.394)	(45.183)	(34.443)	(277.083)
Constant	12.061^{*}	259.584	566.378^{*}	−186.784	1050.288
	(6.654)	(337.060)	(333.569)	(193.252)	(1240.562)
$\ln(\alpha)$	2.367^{***}				
	(0.132)				
Observations	333	333	283	255	55
Log Likelihood	−454.59	−2415.02	−1988.07	−1773.93	−420.06

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure B.3: The Marginal Effect of 2016 Trump Vote
—Full Range of Tribal Population—

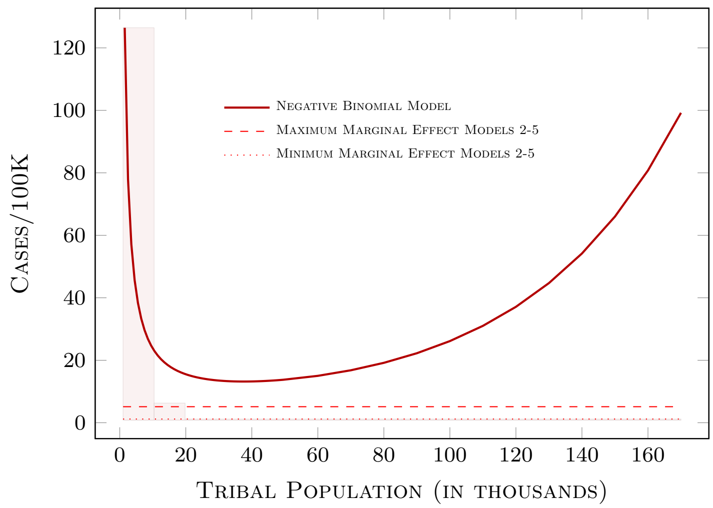


Figure B.4: The Marginal Effect of 2016 Trump Vote
—Tribal Population Between 2000 & 50000 Only—

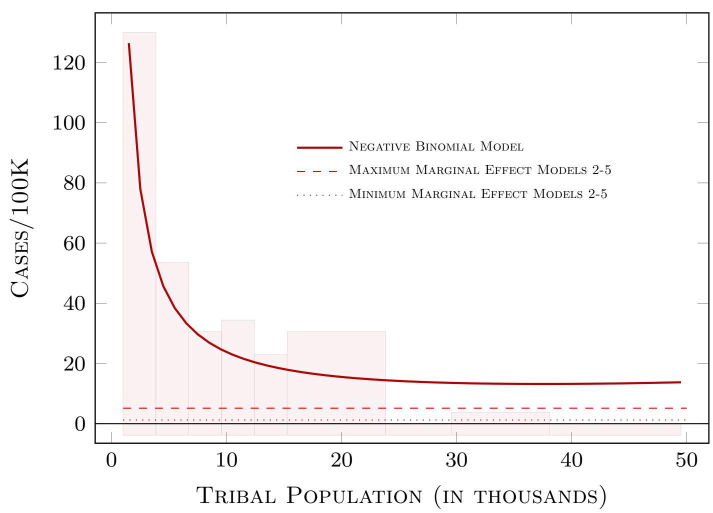


Table B.3: Covid Rate & Republican Governors
—Controlling for Tribal Population—

	(1)	(2)	(3)	(4)	(5)
Republican Governor	1.304**	172.582***	136.967***	109.796***	133.461*
	(0.518)	(31.129)	(33.214)	(31.854)	(72.630)
Tribal Population	0.000	0.012**	0.007**	0.024**	0.001
	(0.000)	(0.005)	(0.003)	(0.010)	(0.003)
% Native on Tribe's Lands	−2.727*	357.169***	207.152***	191.570***	−639.968
	(1.529)	(44.426)	(50.781)	(47.167)	(625.491)
% English only Households	−5.083***	−846.303***	−365.434***	−276.560***	−623.889
	(0.891)	(70.010)	(99.647)	(96.433)	(508.614)
% Households w/Plumbing	−11.088*	149.743	−186.305	−175.065	−476.420
	(6.623)	(187.551)	(229.841)	(186.519)	(1683.614)
Median Age	−0.050	1.835***	0.307	0.497	−10.476
	(0.041)	(0.620)	(0.686)	(0.604)	(14.508)
Median Household Income	0.000	0.001***	0.001***	0.001*	−0.012
	(0.000)	(0.000)	(0.000)	(0.000)	(0.012)
Cases per 100k in State	0.001	0.008	0.008	−0.007	−0.056
	(0.001)	(0.016)	(0.017)	(0.019)	(0.079)
Log Casino Sq. Footage	0.267***	11.535***	8.509***	6.122***	3.095
	(0.044)	(1.714)	(1.830)	(1.756)	(17.077)
Rate of Racial Misclassification	−2.536*	−152.058***	−24.044	18.803	−362.569
	(1.440)	(27.552)	(31.115)	(32.680)	(451.185)
Health Facility on Tribal Land	2.844***	69.456**	57.029**	39.519	−75.085
	(0.850)	(27.069)	(27.822)	(25.769)	(124.061)
State Recognized Tribe	2.390*	314.628***	140.133**	102.332**	−108.611
	(1.324)	(44.192)	(54.669)	(48.203)	(368.531)
Phoenix Indian Health Board	−0.502	−109.355***	192.417***	151.113**	136.568
	(0.508)	(41.490)	(69.410)	(69.639)	(413.873)
Constant	15.502**	292.817	224.020	132.931	2754.127
	(6.822)	(180.149)	(209.950)	(162.243)	(2620.310)
$\ln(\alpha)$	2.350***				
	(0.134)				
Observations	333	333	283	255	55
Log Likelihood	−454.11	−2447.12	−2020.88	−1820.79	−442.59

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure B.5: The Effect of Having a Republican Governor
—Full Range of Tribal Population—

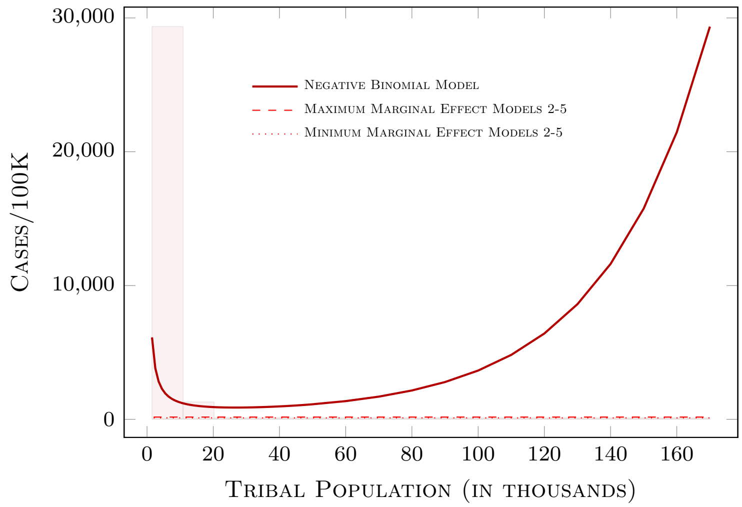
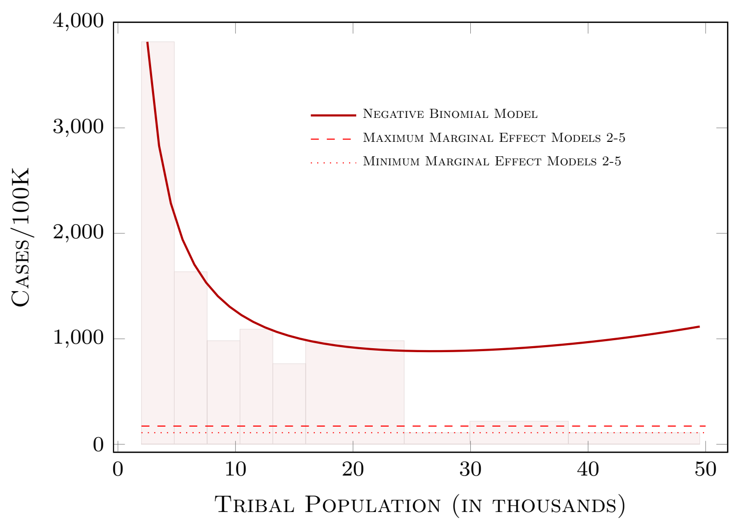


Figure B.6: The Effect of Having a Republican Governor
—Tribal Population Between 2000 & 50000 Only—



C Heckman Selection Models

Most of the observations in Foxworth et al.'s (2022) data appear to have no Covid cases. In fact, after excluding tribes that cross state boundaries as discussed in the manuscript, there are only 55 tribes that are reported as having one more Covid cases. It is not clear whether the observation coded as having no Covid cases i) actually had no Covid cases, ii) if no Covid cases were recorded because little testing took place, or iii) if it was due to challenges in collecting the data.[5] This may be problematic as if the latter two reasons apply, the zeros recorded in the data do not actually represent observations with zero Covid cases and, in effect, those represent missing data. If the missing data is missing at random (or, at least, the missingness is uncorrelated with the model's covariates), the consequences are not too serious. If, however, the same variables that affect the spread of Covid influence whether cases are recorded or not, sample selection bias is introduced. That is a reasonable possibility here. If say, partisanship, affects health outcomes because of the undersupply of health care then it seems plausible that partisanship affects testing for Covid as well as the infrastructure that may facility reporting of data.

An approach to address this problem is to model the sample selection along with the Covid rate (when reported) with the Heckman selection model. The Heckman model basically consists of two models, a selection equation and an outcome equation. One challenge in estimating the model is that the selection equation must include at least one covariate that is not included in the outcome equation. Thus, I have excluded several variables from the outcome equation that seemed more likely to influence the selection than the outcome (i.e., the Covid rate). The decision of which variables to exclude is somewhat subjective but the interested reader can experiment with the models provided in the replication files.

The results are presented in table C.4. The results are instructive. The partisanship variables have a statistically significant effect in the selection equation but do not appear to matter in the outcome equation. That is, when accounting for selection, it does not appear that partisanship affects the Covid rate (where at least one case is reported). This casts some doubt on the results reported in this paper (and, of course, in Foxworth et al. (2022) although there are additional reasons to question those results as discussed above). However, at the end of the

[5]The data collected consisted of crowd-sourced data as well as information gathered from publicly released information by tribes.

day, we lack information to be able to conclusively say which results are accurate. That is, if it was the case that observations with no Covid cases reported did indeed have no Covid cases, the application of the Heckman model would not be warranted. On the face of it, that seems a little unlikely.

Another interesting result is that the effect of partisanship in the selection equation is positive, indicating that tribes in more Republican states are more likely to have at least one Covid case. That goes against the idea that the provision of health services on Native American lands, including testing services, was lacking in Republican areas. It is, however, consistent with the argument that more Republican areas produce worse outcomes — i.e., it is more likely that there was at least one case — but that, in turn, is contradicted by the fact that partisanship is not statistically significant in the outcome equation. In sum, the Heckman results raise questions whose answers are likely to remain unanswered with the data available. In order to adequately address these questions, more detailed data about whether the absence of Covid cases in a given tribe means that there actually were no cases or if the information is missing due to lack or testing or challenges in collecting the data.

Table C.4: Heckman Selection Models

	(1)	(2)
% Trump vote in 2016	11.626	
	(30.985)	
Republican Governor		270.214
		(671.788)
% English only Households	−2917.179***	−2867.733***
	(1112.690)	(1059.699)
% Households w/Plumbing	4104.716	3994.305
	(4957.624)	(4932.501)
Rate of Racial Misclassification	−1015.483	−920.992
	(2220.660)	(2232.068)
Health Facility on Tribal Land	−722.855	−739.563
	(1223.234)	(1212.553)
State Recognized Tribe	−121.537	343.928
	(2085.867)	(2121.785)
Phoenix Indian Health Board	−443.423	−668.963
	(852.122)	(937.569)
Constant	1039.296	1525.288
	(5863.670)	(5593.147)
% Trump vote in 2016	0.029***	
	(0.010)	
Republican Governor		0.681***
		(0.228)
Tribal Population	0.000***	0.000***
	(0.000)	(0.000)
% English only Households	−1.019**	−0.749
	(0.509)	(0.488)
% Households w/Plumbing	−2.311	−2.637
	(2.011)	(1.978)
Median Age	−0.002	−0.002
	(0.014)	(0.014)
Median Household Income	0.000	0.000
	(0.000)	(0.000)
Log Casino Sq. Footage	0.064***	0.062***
	(0.023)	(0.023)
Rate of Racial Misclassification	−2.396***	−2.589***
	(0.815)	(0.822)
Health Facility on Tribal Land	0.349	0.344
	(0.369)	(0.366)
State Recognized Tribe	−0.473	−0.321
	(0.810)	(0.755)
Phoenix Indian Health Board	−0.175	−0.480
	(0.364)	(0.402)
Constant	2.443	4.057*
	(2.280)	(2.178)
atanhρ	−0.430	−0.455
	(0.327)	(0.323)
ln(σ)	7.523***	7.526***
	(0.126)	(0.127)
Observations	283	283
Log Likelihood	−588.50	−588.25

D State-Tribe Differences in Covid Rate

The discussion of partisanship and political ideology in Foxworth et al. (2022, p. 5) highlights how partisanship and ideology shaped responses to the pandemic but does not explain why these factors should affect Native Americans more, or differently, than the general population in the states. It is, however, fairly clear that Foxworth et al.'s (2022) interest is in exploring whether more Republican states (along with other factors) treat Native Americans differently than they do other residents. If that interpretation is correct, the Covid rate in the state as a whole is a useful benchmark against which to measure how members of the Native Nations are affected. While Foxworth et al. (2022) do control for the state-level Covid rate, this can be captured more directly by using the difference between each tribe's Covid rate and the statewide Covid rate as the dependent variable, i.e., it would capture how much worse (or better) Native Americans fare compared to the state as a whole.

The results suggest Native Americans experience higher Covid rates relative to residents in the state where their tribal lands are. Tables D.6 & D.5 show that the estimated effect is substantial. For example, a ten-point increase in Trump support in the 2016 election corresponds to 62-103 additional Covid cases per 100k inhabitants (depending on which observations are included) while a Republican governor translates into 290-361 additional cases per 100k. Table D.7 presents the results of Heckman selection models that suggest that the Covid rate differential is unlikely to be influenced by partisanship while the partisanship coefficients have a statistically significant effect in the selection part of the model, i.e., whether at least one Covid case was recorded. This is in line with the findings of the selection models discussed above and, thus, raises important questions about how partisanship affects the observed, or measured, Covid rate. That is, is it because Native Americans are more likely to contract Covid in more Republican-leaning states? Or is it because cases are more likely to be detected?

Table D.5: TRUMP SUPPORT & THE COVID RATE DIFFERENTIAL
—STATE VS. TRIBE—

	(1) ORIG. OBS.	(2) UNIQUE OBS.	(3) POP.<20K	(4) >0 CASES
% TRUMP VOTE IN 2016	7.630***	10.339***	6.989***	6.220**
	(0.668)	(0.826)	(0.959)	(2.485)
TRIBAL POPULATION	0.020***	0.005**	0.034***	0.003*
	(0.000)	(0.002)	(0.005)	(0.002)
% NATIVE ON TRIBE'S LANDS	182.939***	231.271***	110.675***	−388.526*
	(25.815)	(28.430)	(34.022)	(207.952)
% ENGLISH ONLY HOUSEHOLDS	−623.378***	−501.179***	−391.335***	−168.864
	(55.530)	(58.494)	(64.070)	(290.580)
% HOUSEHOLDS W/PLUMBING	−120.822	−349.279***	17.291	−1080.212*
	(104.200)	(131.809)	(134.517)	(628.401)
MEDIAN AGE	0.989*	−0.776	−0.474	−16.739**
	(0.507)	(0.583)	(0.618)	(7.124)
MEDIAN HOUSEHOLD INCOME	0.000	0.001***	0.000	−0.005
	(0.000)	(0.000)	(0.000)	(0.003)
LOG CASINO SQ. FOOTAGE	9.037***	10.951***	5.322***	2.973
	(1.177)	(1.189)	(1.604)	(6.498)
RATE OF RACIAL MISCLASSIFICATION	−274.521***	−143.247***	−184.711***	−651.032***
	(37.625)	(41.079)	(46.098)	(233.126)
HEALTH FACILITY ON TRIBAL LAND	85.691***	126.390***	88.369***	−324.373***
	(29.556)	(32.083)	(27.964)	(92.055)
STATE RECOGNIZED TRIBE	20.605	21.290	−49.158	−974.996***
	(37.810)	(37.072)	(32.317)	(127.568)
PHOENIX INDIAN HEALTH BOARD	19.044	115.787***	56.811**	161.517
	(25.678)	(27.487)	(25.442)	(158.675)
CONSTANT	−3.579	−189.160	−294.562**	2671.077**
	(130.852)	(122.869)	(120.841)	(991.414)
OBSERVATIONS	333	283	255	55
LOG LIKELIHOOD	−2365.16	−1950.93	−1748.21	−414.50

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table D.6: Republican Governors & the Covid Rate Differential
—State vs. Tribe—

	(1) Orig. Obs.	(2) Unique Obs.	(3) Pop.<20K	(4) >0 cases
Republican Governor	361.075***	301.672***	280.806***	280.147***
	(20.586)	(19.186)	(17.192)	(81.728)
Tribal Population	0.010***	0.006***	0.019***	0.004
	(0.002)	(0.001)	(0.003)	(0.004)
% Native on Tribe's Lands	289.875***	162.686***	141.018***	−45.467
	(36.508)	(35.574)	(35.529)	(341.257)
% English only Households	−714.793***	−348.206***	−320.029***	−138.402
	(64.996)	(67.160)	(64.507)	(415.311)
% Households w/Plumbing	81.453	−111.476	270.716***	−490.650
	(103.934)	(95.560)	(97.794)	(700.623)
Median Age	2.253***	−0.344	−0.548	−8.325
	(0.649)	(0.718)	(0.512)	(11.164)
Median Household Income	0.001**	0.000	0.000	−0.002
	(0.000)	(0.000)	(0.000)	(0.006)
Log Casino Sq. Footage	7.004***	3.082**	0.704	−0.492
	(1.506)	(1.459)	(1.442)	(10.171)
Rate of Racial Misclassification	−444.618***	−271.726***	−244.382***	−593.240
	(36.705)	(44.347)	(37.339)	(464.194)
Health Facility on Tribal Land	51.644**	26.052	36.837	−297.189***
	(20.347)	(35.918)	(31.497)	(101.528)
State Recognized Tribe	−6.188	−145.384***	−132.822**	−715.335***
	(47.504)	(51.400)	(52.014)	(202.580)
Phoenix Indian Health Board	−191.864***	−57.322	−46.750	37.959
	(39.406)	(36.478)	(30.821)	(213.829)
Constant	270.539**	203.779	−188.384*	1660.933
	(122.235)	(127.030)	(106.032)	(1867.753)
Observations	333	283	255	55
Log Likelihood	−2395.18	−1965.52	−1769.61	−427.18

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table D.7: Covid Rate Differential
—Heckman Selection Models—

	(1)	(2)
% Trump vote in 2016	13.378	
	(31.783)	
Republican Governor		312.162
		(687.917)
% English only Households	−2649.960**	−2591.463**
	(1143.992)	(1089.181)
% Households w/Plumbing	2690.347	2555.507
	(5094.951)	(5067.885)
Rate of Racial Misclassification	−959.522	−854.502
	(2273.592)	(2284.763)
Health Facility on Tribal Land	−909.620	−929.312
	(1255.550)	(1243.347)
State Recognized Tribe	−746.465	−219.042
	(2138.430)	(2173.683)
Phoenix Indian Health Board	−476.463	−736.657
	(875.383)	(963.655)
Constant	1821.699	2396.790
	(6033.154)	(5747.524)
% Trump vote in 2016	0.029***	
	(0.010)	
Republican Governor		0.682***
		(0.229)
Tribal Population	0.000***	0.000***
	(0.000)	(0.000)
% English only Households	−1.016**	−0.746
	(0.509)	(0.488)
% Households w/Plumbing	−2.349	−2.681
	(2.016)	(1.984)
Median Age	−0.002	−0.002
	(0.014)	(0.014)
Median Household Income	0.000	0.000
	(0.000)	(0.000)
Log Casino Sq. Footage	0.064***	0.062***
	(0.023)	(0.023)
Rate of Racial Misclassification	−2.405***	−2.601***
	(0.815)	(0.823)
Health Facility on Tribal Land	0.348	0.343
	(0.369)	(0.365)
State Recognized Tribe	−0.480	−0.328
	(0.811)	(0.755)
Phoenix Indian Health Board	−0.174	−0.479
	(0.363)	(0.402)
Constant	2.484	4.107*
	(2.284)	(2.185)
atanhρ	−0.454	−0.485
	(0.322)	(0.319)
ln(σ)	7.554***	7.558***
	(0.128)	(0.130)
Observations	283	283

E Heteroscedasticity & the Log the Covid Rate

The textbook approach to address heteroscedasticity involves deriving a more efficient estimator. Instead of adopting this standard approach, Foxworth et al. (2022) use the log of the Covid rate (cases per 100k inhabitants) as their dependent variable.[6] While taking the log of a variable can help with heteroscedasticity, it involves an assumption about the nature of the relationship between the dependent and independent variables. That is, the relationship between the Covid rate and the independent variables is no longer assumed to be linear. This implies, for example, that the effect of Trump's vote share is not constant across tribes, but it is not clear why an increase in the effect of Trump support should differ from tribe to tribe.

To demonstrate how Foxworth et al.'s (2022) model specification implies that the effect of the covariates is proportional to the Covid rate, suppose we estimate the following model:

$$\ln(r) = \beta_0 + \beta_p P + \beta_z Z) \tag{1}$$

where r is the Covid rate, P is a measure of partisanship, and Z represents other covariates.[7] Then β_p is the marginal effect of partisanship on the log of the Covid rate. As the model is not linear in partisanship, the effect of partisanship on the Covid Rate is not constant. Thus, in order to interpret the results in terms of a change in the Covid Rate, which is the quantity of interest, we must ask what the effect of partisanship at different 'initial' Covid rates. The (marginal) effect of Covid on partisanship is simply the derivative of the Covid rate with respect to partisanship. Thus, we take the natural exponent of both sides of equation 1:

$$r \quad = \quad e^{(\beta_0 + \beta_p P + \beta_z Z)} \tag{2}$$

Then, we obtain the marginal effect of P is the derivative of the RHS w.r.t. P using the chain rule,[8] or:

[6] As there are observations with zero Covid cases, they add a constant to the Covid rate to avoid losing observations.

[7] Foxworth et al. (2022) estimate a model where the dependent variable is $\ln(r+1)$ as there are many observations where $r = 0$ and $\ln(0)$ is not defined. It should be clear that not much changes in terms of the marginal effects whether we use r and $r+1$ and, thus, for simplicity, I simply use r.

[8] That is, $\frac{d}{dx}e^x = e^x$ and $\frac{d}{dx}e^u = e^x \frac{du}{dx}$ so, e.g., $\frac{d}{dx}e^{\beta x} = \beta e^{\beta x}$

$$\frac{\partial r}{\partial P} = \beta_p[e^{\beta_0}e^{\beta_p P}e^{\beta_z Z}] \tag{3}$$

That is, in the model, the estimated marginal effect of partisanship on the Covid rate is the coefficient on partisanship, β_P times the expected Covid rate for the tribe (based on the other covariates in the model). Table E.8 displays the results of the regression models as provided in Foxworth et al.'s (2022) supplementary materials. Given the above, we can now give an interpretation to the effects of the partisanship variables in terms of the Covid rate (rather than the log of the Covid rate). Starting with Trump's support in the 2016 election we can see that the marginal effect of an increase in his vote share would be 5 cases per 100k for a tribe whose estimated Covid rate is 175 cases per 100k (5^{th} percentile of tribes reporting at least one Covid case) but it would be estimated to be 28 cases per 100k for a tribe with estimated rate of 954 cases per 100k (95^{th} percentile). Thus, the effect of partisanship is over five times as great in the latter case. Importantly, these differences in the size of the effect are not driven by patterns in the data but by the use of a model that explicitly models the marginal effect of the variable as a function of the expected value of the dependent variable.

If we consider the estimated effect of having a Republican governor, we naturally get similar differences. Considering again the number of cases at the 5th and 95th percentiles we find that in the former having a Republican governor is associated with an increase of 168 cases per 100k for the former tribe and 904 cases per 100k for the latter. Not only is the difference between the two scenarios very different, it is also very large. A coefficient of .95 implies that having a Republican Governor very nearly doubles the Covid rate.

Table E.8: Regression with Log of Covid Rate (+1)

	(1)	(2)
% Trump vote in 2016	0.030**	
	(0.012)	
Republican Governor		0.948***
		(0.295)
% English only Households	−2.649***	−2.562***
	(0.764)	(0.773)
% Households w/Plumbing	−5.067*	−4.719*
	(2.821)	(2.784)
Median Age	0.001	0.005
	(0.013)	(0.012)
Median Household Income	0.000	0.000
	(0.000)	(0.000)
Cases per 100k in State	0.001*	0.001*
	(0.000)	(0.000)
Log Casino Sq. Footage	0.093***	0.093***
	(0.025)	(0.024)
Rate of Racial Misclassification	−0.941*	−1.037**
	(0.504)	(0.509)
Health Facility on Tribal Land	0.892**	0.883**
	(0.359)	(0.355)
State Recognized Tribe	0.773*	0.869**
	(0.454)	(0.439)
Phoenix Indian Health Board	0.579	0.394
	(0.471)	(0.457)
Constant	5.876**	6.572***
	(2.686)	(2.528)
Observations	333	333
Log Likelihood	−732.59	−729.76

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$